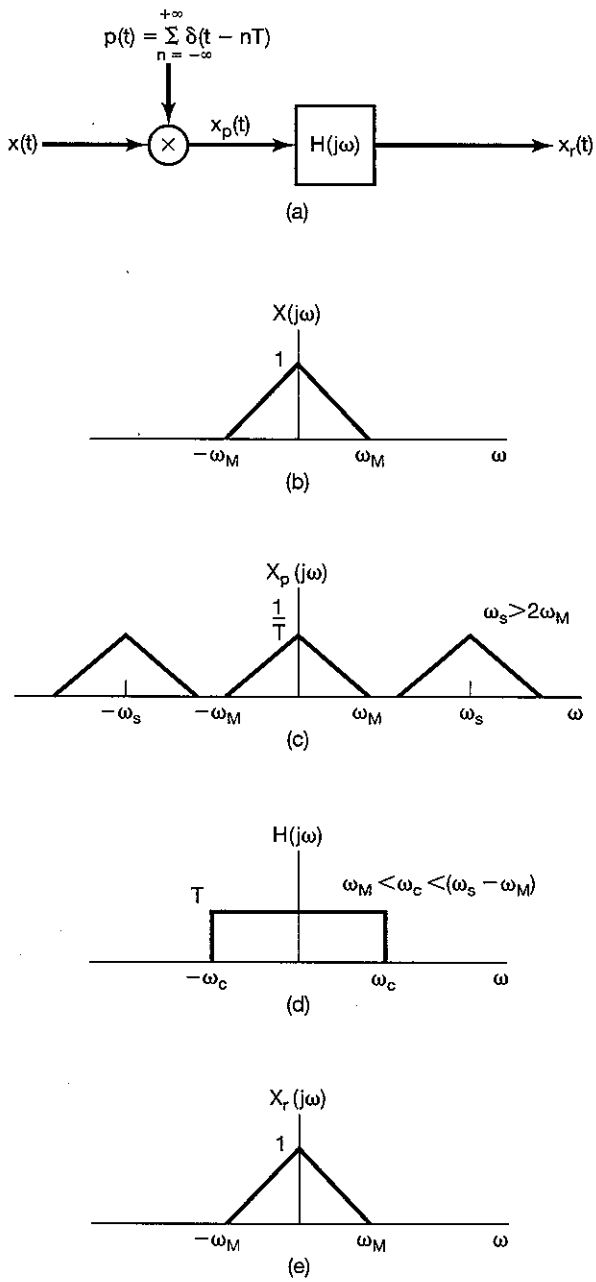


Figure 7.4 Exact recovery of a continuous-time signal from its samples using an ideal lowpass filter: (a) system for sampling and reconstruction; (b) representative spectrum for $x(t)$; (c) corresponding spectrum for $x_p(t)$; (d) ideal lowpass filter to recover $X(j\omega)$ from $X_p(j\omega)$; (e) spectrum of $x_r(t)$.

The frequency $2\omega_M$, which, under the sampling theorem, must be exceeded by the sampling frequency, is commonly referred to as the *Nyquist rate.*[2]

As discussed in Chapter 6, ideal filters are generally not used in practice for a variety of reasons. In any practical application, the ideal lowpass filter in Figure 7.4 would be

[2]The frequency ω_M corresponding to one-half the Nyquist rate is often referred to as the *Nyquist frequency.*

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n}, \tag{7.19}$$

or, using eq. (7.16),

$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n}. \tag{7.20}$$

Comparing eqs. (7.18) and (7.20), we see that $X_d(e^{j\Omega})$ and $X_p(j\omega)$ are related through

$$X_d(e^{j\Omega}) = X_p(j\Omega/T). \tag{7.21}$$

Also, recall that, as developed in eq. (7.6) and illustrated in Figure 7.3,

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)). \tag{7.22}$$

Consequently,

$$X_d(e^{j\Omega}) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\Omega - 2\pi k)/T). \tag{7.23}$$

The relationship among $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ is illustrated in Figure 7.22 for two different sampling rates. Here, $X_d(e^{j\Omega})$ is a frequency-scaled version of $X_p(j\omega)$

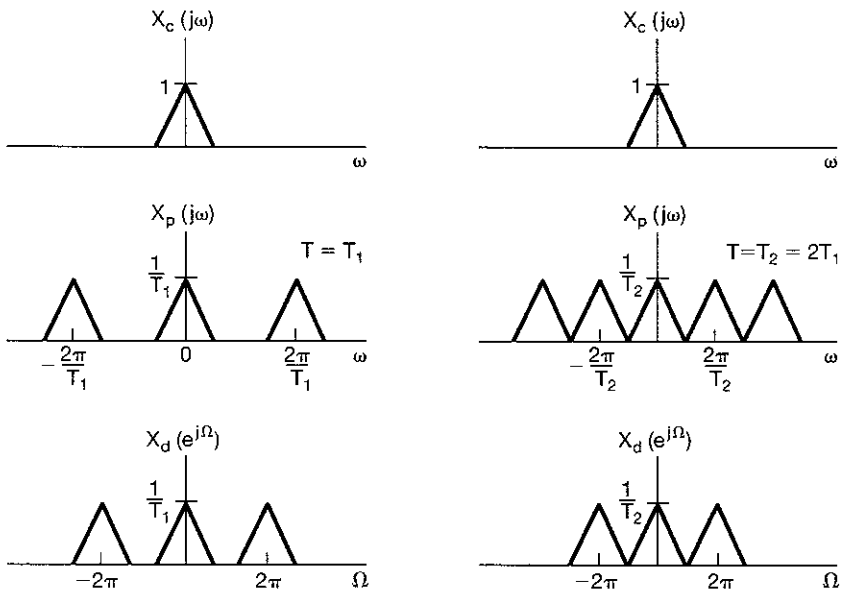


Figure 7.22 Relationship between $X_c(j\omega)$, $X_p(j\omega)$, and $X_d(e^{j\Omega})$ for two different sampling rates.

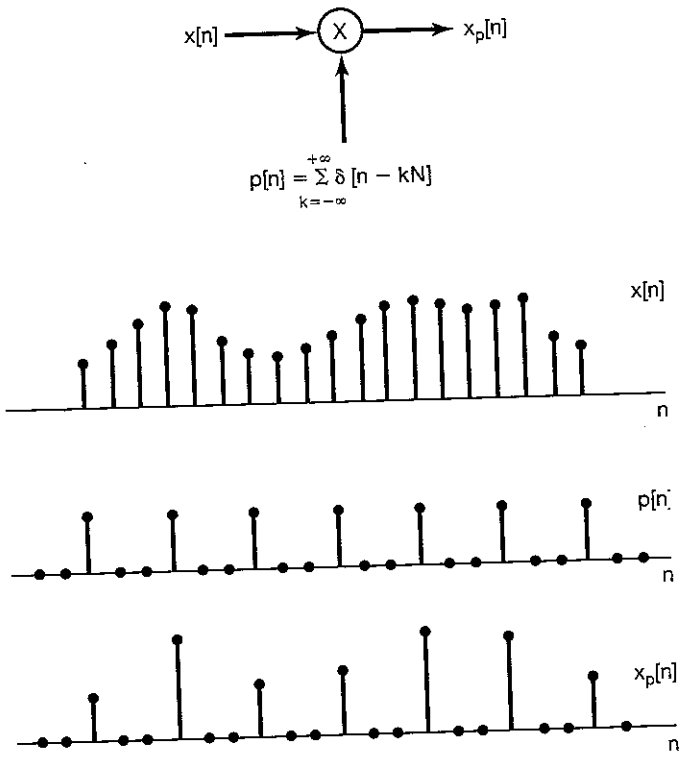


Figure 7.31 Discrete-time sampling.

As in Example 5.6, the Fourier transform of the sampling sequence $p[n]$ is

$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s),$$

where ω_s, the sampling frequency, equals $2\pi/N$. Combining eqs. (7.40) and (7 have

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X\left(e^{j(\omega - k\omega_s)}\right).$$

Equation (7.42) is the counterpart for discrete-time sampling of eq continuous-time sampling and is illustrated in Figure 7.32. In Figure 7.32 $\omega_s - \omega_M > \omega_M$, or equivalently, $\omega_s > 2\omega_M$, there is no aliasing [i.e., the nonzer of the replicas of $X(e^{j\omega})$ do not overlap], whereas with $\omega_s < 2\omega_M$, as in Figur frequency-domain aliasing results. In the absence of aliasing, $X(e^{j\omega})$ is faithfu duced around $\omega = 0$ and integer multiples of 2π. Consequently, $x[n]$ can be from $x_p[n]$ by means of a lowpass filter with gain N and a cutoff frequency gr

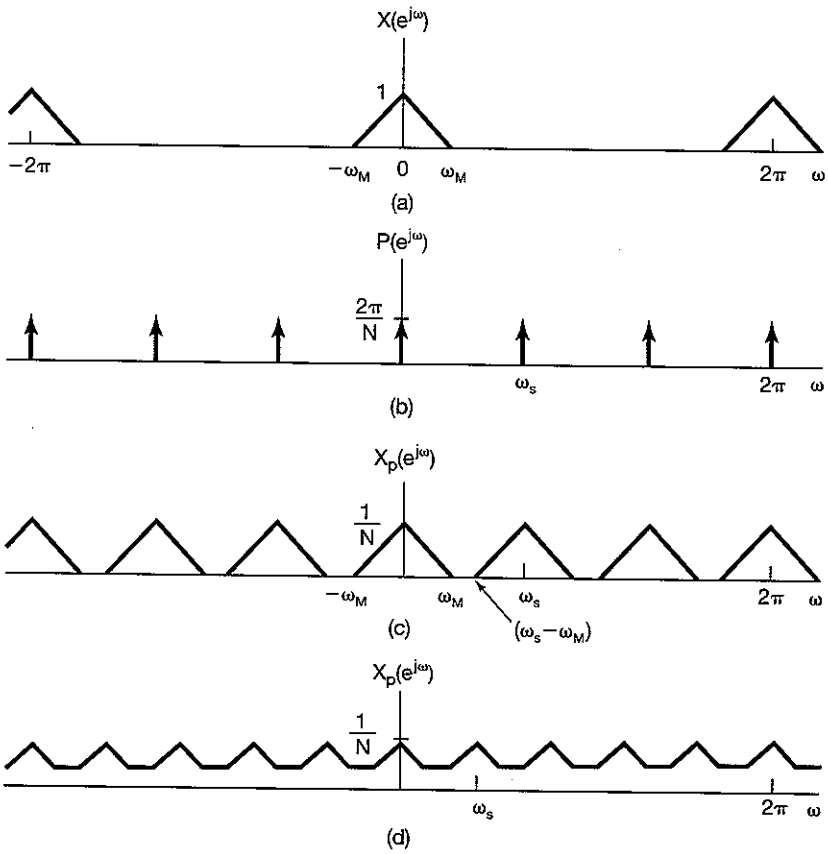


Figure 7.32 Effect in the frequency domain of impulse-train sampling of a discrete-time signal: (a) spectrum of original signal; (b) spectrum of sampling sequence; (c) spectrum of sampled signal with $\omega_s > 2\omega_M$; (d) spectrum of sampled signal with $\omega_s < 2\omega_M$. Note that aliasing occurs.

ω_M and less than $\omega_s - \omega_M$, as illustrated in Figure 7.33, where we have specified the cutoff frequency of the lowpass filter as $\omega_s/2$. If the overall system of Figure 7.33(a) is applied to a sequence for which $\omega_s < 2\omega_M$, so that aliasing results, $x_r[n]$ will no longer be equal to $x[n]$. However, as with continuous-time sampling, the two sequences *will* be equal at multiples of the sampling period; that is, corresponding to eq. (7.13), we have

$$x_r[kN] = x[kN], \quad k = 0, \pm 1, \pm 2, \ldots, \tag{7.43}$$

independently of whether aliasing occurs. (See Problem 7.46.)

Furthermore, we recognize the right-hand side of eq. (7.52) as the Fourier transform of $x_p[n]$; that is,

$$\sum_{n=-\infty}^{+\infty} x_p[n]e^{-j\omega n/N} = X_p(e^{j\omega/N}). \tag{7.53}$$

Thus, from eqs. (7.52) and (7.53), we conclude that

$$X_b(e^{j\omega}) = X_p(e^{j\omega/N}). \tag{7.54}$$

This relationship is illustrated in Figure 7.35, and from it, we observe that the spectra for the sampled sequence and the decimated sequence differ only in a frequency scaling or normalization. If the original spectrum $X(e^{j\omega})$ is appropriately band limited, so that there is no aliasing present in $X_p(e^{j\omega})$, then, as shown in the figure, the effect of decimation is to spread the spectrum of the original sequence over a larger portion of the frequency band.

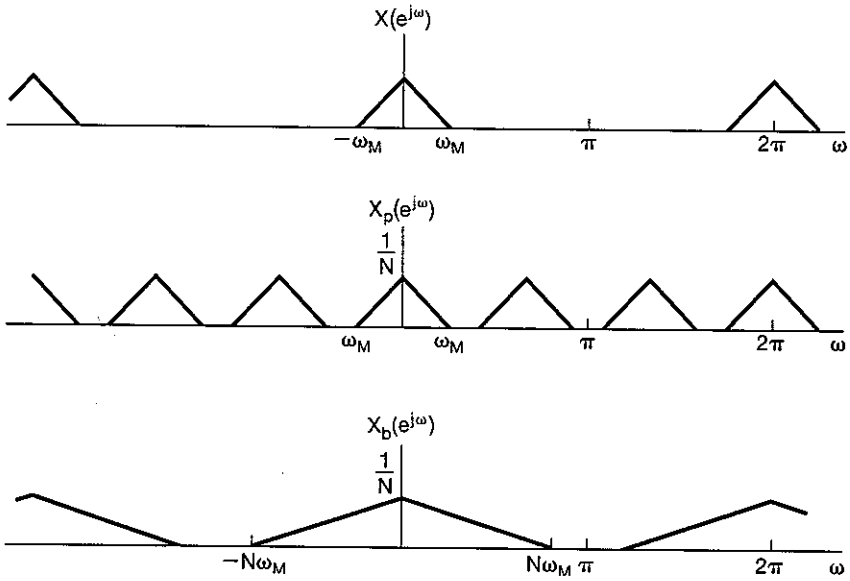


Figure 7.35 Frequency-domain illustration of the relationship between sampling and decimation.

If the original sequence $x[n]$ is obtained by sampling a continuous-time signal, the process of decimation can be viewed as reducing the sampling rate on the signal by a factor of N. To avoid aliasing, $X(e^{j\omega})$ cannot occupy the full frequency band. In other words, if the signal can be decimated without introducing aliasing, then the original continuous-time signal was oversampled, and thus, the sampling rate can be reduced without aliasing. With the interpretation of the sequence $x[n]$ as samples of a continuous-time signal, the process of decimation is often referred to as *downsampling*.